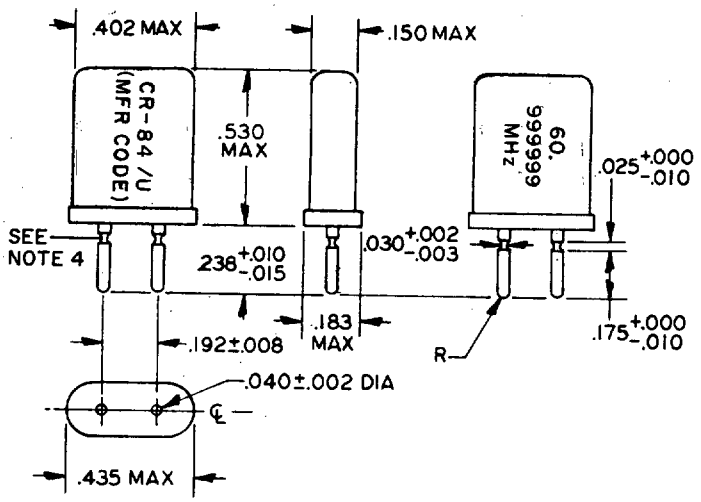
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-84/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics: 17 to 61 MHz; third overtone; controlled; series.



INCHES	MM	INCHES	MM
.002	.05	.150	3.81
.003	.08	.175	4.45
.008	.20	.183	4.65
.010	.25	.192	4.88
.015	.38	.238	6.05
.025	.64	.402	10.21
.030	.76	.435	11.05
.040	1.02	.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designation, frequency, and code designating characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.
4. The pin undercut may be omitted.

(F) denotes change

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-25/U (MIL-H-10056/11)
Dimensions, marking, and configuration	See figure
Frequency range	17 to 61 MHz, inclusive
Capacitance, shunt	7 pF, maximum
Frequency tolerances:	
Operating temperature range	±20 PPM
Room temperature	±70 PPM
Frequency stability	± 5 PPM
Equivalent resistance	40 ohms, maximum
Mode of oscillation	Third mechanical overtone
Test set, calibration values, and rated drive level	See table II
Reference temperature	85° ±1° C
Temperature ranges:	
Operable	-55° to +80° C and +90° to +105° C, inclusive
(F) Operating	80° to 90° C, inclusive
Shock (specified pulse):	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
(F) Thermal shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	5 PPM

1/ Method 204 of MIL-STD-202, test condition A.

NOTE: NATO guidance type NXT37.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values		Rated drive level
		Resistance	Resistor voltage drop	
	MHz	Ohms	Volt	mW
TS-683/TSM	17 to 25	40	0.28	2.0 ± 0.4
	25+ to 61	40	0.20	1.0 ± 0.2

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-0511-11)

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